Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

Course Title	:	CAD Practice
Course Code	:	ME 2130
Experiment No	:	02
Name of Experiment	:	Application of Basic Solid Modeling Techniques for the Development of Three-Dimensional Parts
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Experiment No: 01

Name of Experiment: Application of Basic Solid Modeling Techniques for the Development of 3D Parts

1.1 Objectives

1. To model the seven assigned parts in SolidWorks using parametric workflows.
2. To produce fully defined 2D sketches and convert them into 3D geometry using features such as Extruded Boss/Base, Revolved Boss/Base, Cut, Fillet, Chamfer, and Shell.
3. To apply feature organization: mates, feature order, mirrors, and patterns to create robust, modifiable parts.
4. To document modeling decisions, common failure modes, and corrective steps for future design work.
5. To compare modeling strategies and justify the choice of features for each part with reference to CAD best practices [1], [2].

1.2 Theory

This lab continues the exploration of parametric solid modeling principles using SolidWorks [1], [3]. Central to parametric modeling is the notion of a fully defined sketch: a 2D profile constrained by dimensions and geometric relations so that subsequent features behave predictably [2].

Common features used in this experiment include:

- Extruded Boss/Base: creates prismatic geometry by giving a 2D profile a linear thickness. - Revolved Boss/Base: generates axisymmetric parts by revolving a profile about a centerline. - Cut operations: remove material from bodies using sketch-based profiles or intersecting geometry. - Fillets and Chamfers: apply local geometry changes to control stress concentrations and assembly fit. - Shell, Mirror, and Pattern features: promote reuse and maintainability of features across a part.

Selecting the correct combination of these tools reduces rebuild time and produces more robust models. For example, thin-walled parts benefit from the 'Shell' feature, while repeating geometry is best handled with 'Pattern' or 'Mirror' features. Fully defining sketches and naming important sketches/features aids later edits and collaboration [1], [2].

Figures in the lab document illustrate representative sketches and feature outcomes from the seven parts modeled in this experiment. Each part's section explains the chosen workflow and demonstrates how the features above were combined to produce the final geometry.

A central advantage of parametric modeling is the capture of design intent: dimensions and constraints express how a part should change when parameters are modified, enabling rapid iteration and variant generation [1]. Good sketching practice therefore emphasizes the use of driving dimensions for functional sizes, and geometric relations (coincident, concentric, tangent, perpendicular, parallel) to lock the sketch topology. Over-constraining a sketch can be as problematic as under-constraining it; the goal is clarity of intent rather than maximal constraint density [2].

Reference geometry (planes, axes, and construction lines) plays an important role when creating complex features like revolved profiles or lofts. Establishing named planes and sketch origins early prevents ambiguous references and reduces rebuild failures when the model history changes. Where repeated variations are required, ‘Configurations‘ or design tables provide a controlled way to generate multiple part variants from a single master model, which is useful for family-of-parts exercises and assembly studies [3].

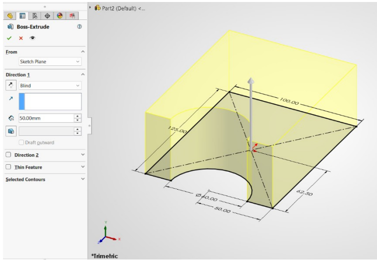
Finally, consideration of manufacturability and downstream processes should influence modeling choices. Simple features and regular faces ease CNC toolpaths and inspection; avoid unnecessarily intricate sketches when a combination of extrudes, cuts, and standard features (fillet/chamfer) will achieve the same form. Following these principles produces parts that are easier to edit, document, and prepare for production workflows [1], [2].

1.3 Apparatus Required

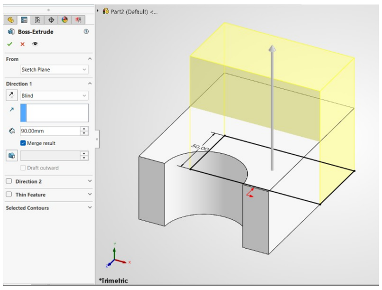
1. **Hardware:** A personal computer (PC) with hardware specifications sufficient for running SolidWorks smoothly.
2. **Software:** An installed and licensed version of the SolidWorks 3D CAD software by Dassault Systèmes.

1.4 Object 1

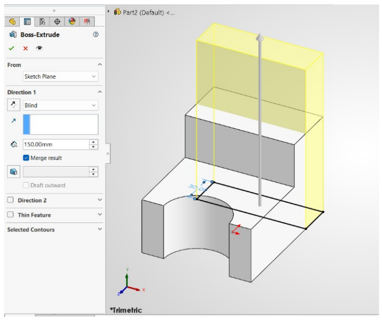
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



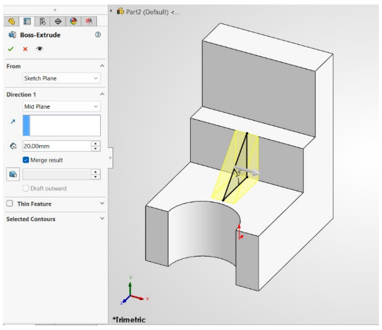
(a) 2D sketch of the base with dimensions



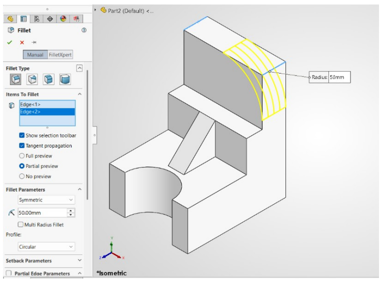
(b) Extruding the base sketch



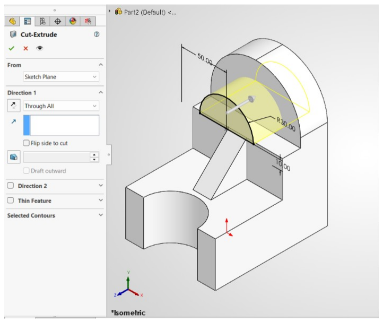
(c) 2D sketch of the top portion with dimensions



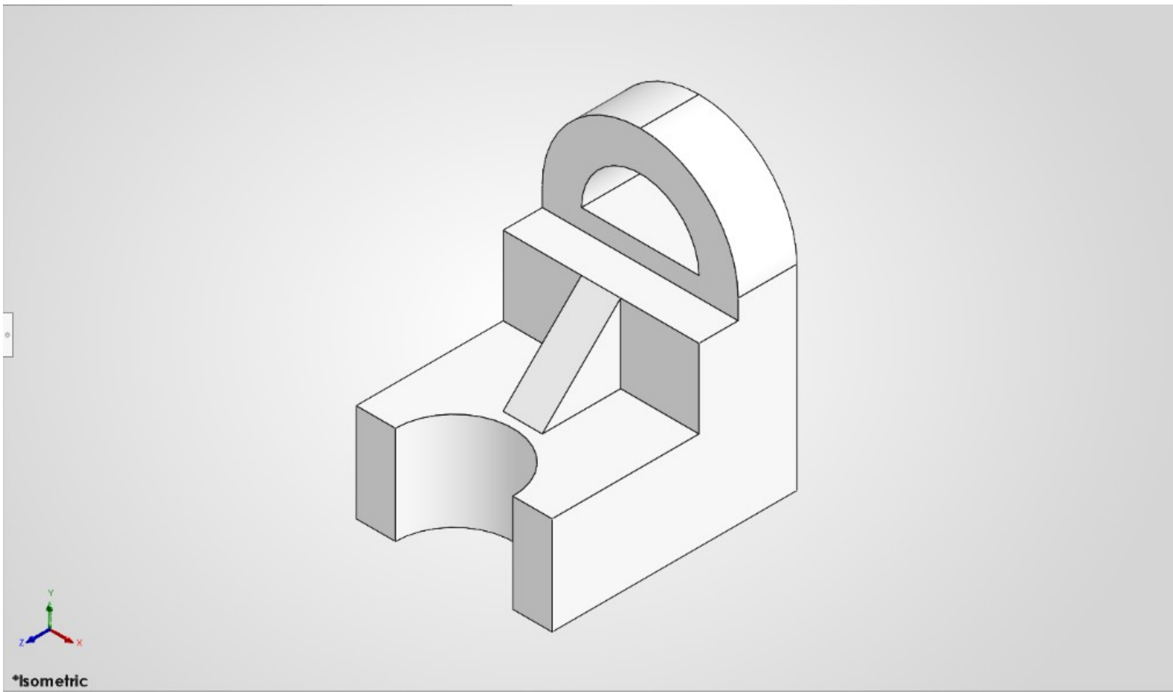
(d) Extruding the top portion



(e) 2D sketch of the top portion with dimensions



(f) Extruding the top portion

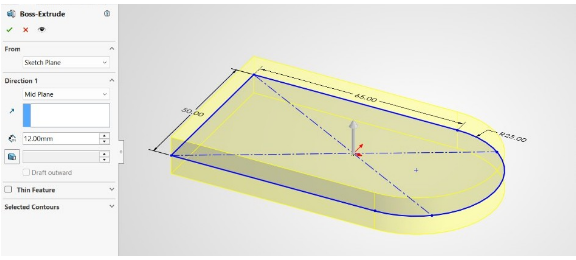


(g) Final result

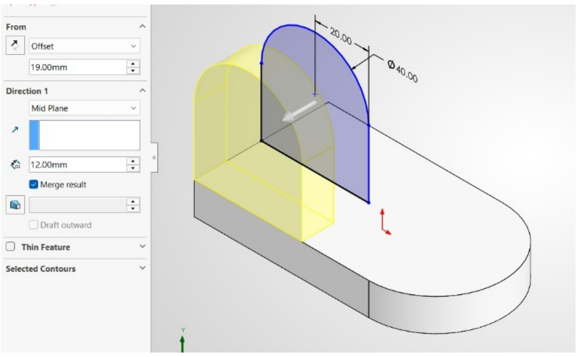
Figure 1.1: Snapshots while Designing Object 1 in Solidworks

1.5 Object 2

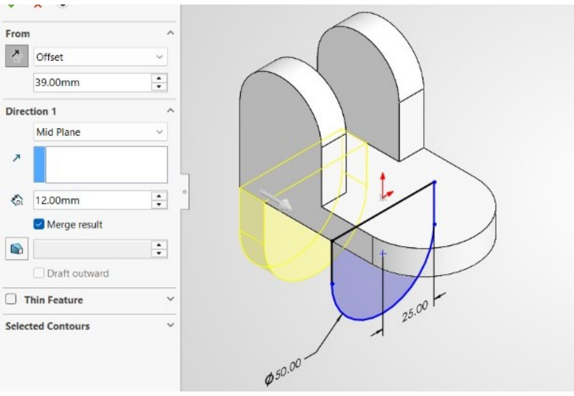
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



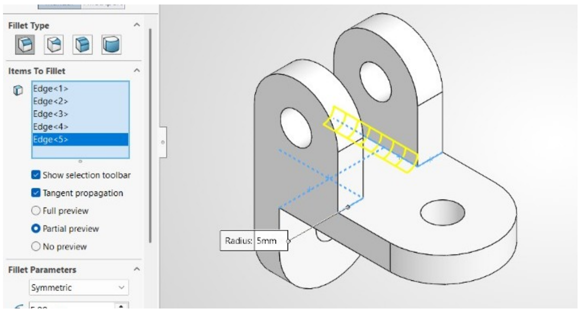
(a) 2D sketch of the base with dimensions



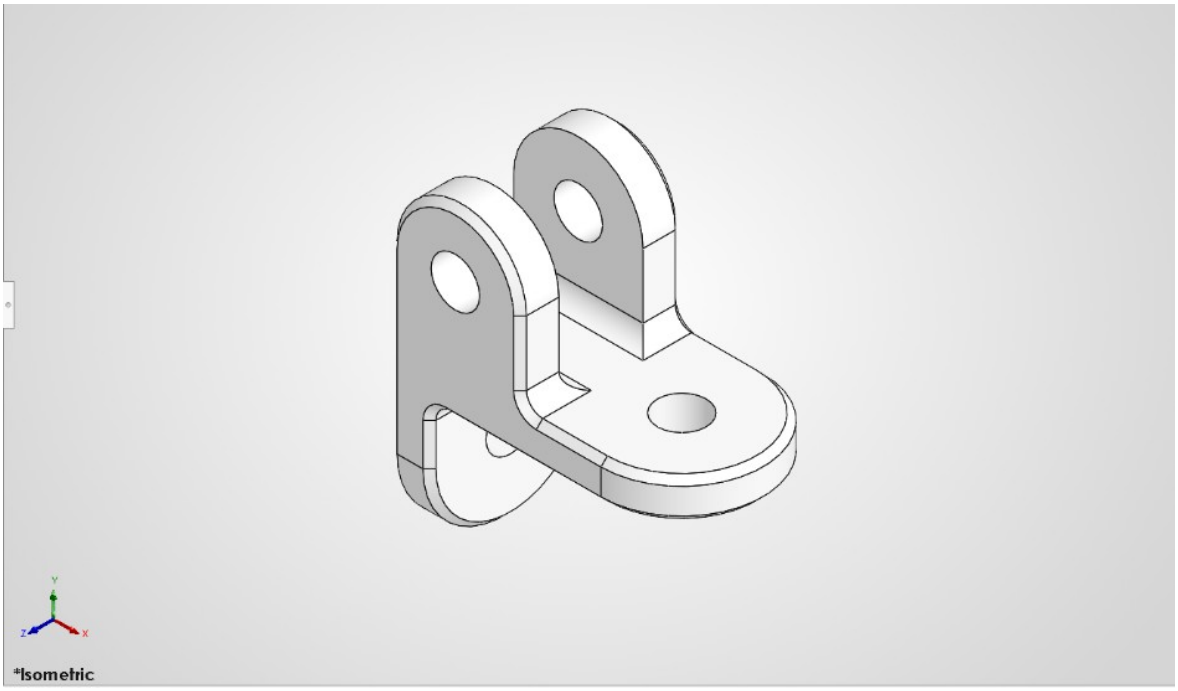
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

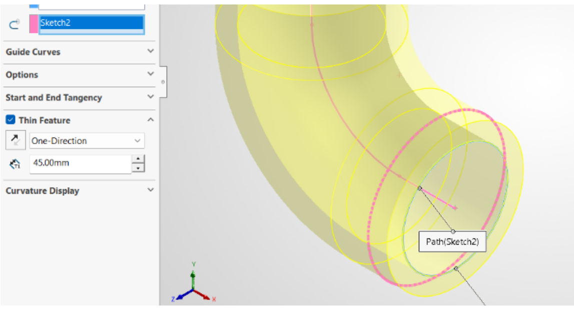


(e) Final result

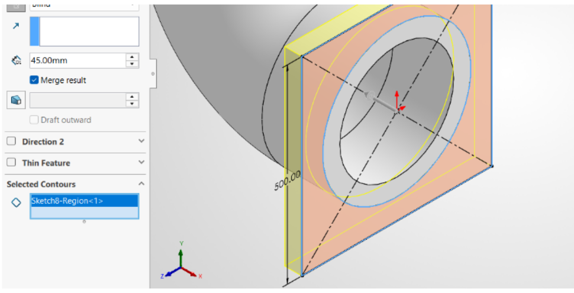
Figure 1.2: Snapshots while Designing Object 2 in Solidworks

1.6 Object 3

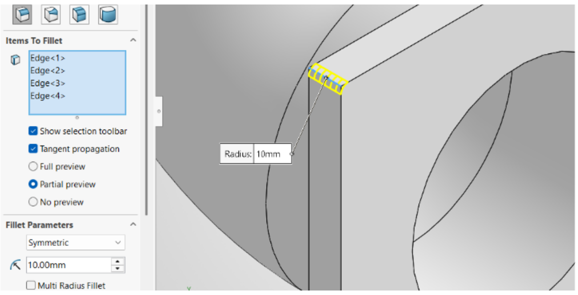
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



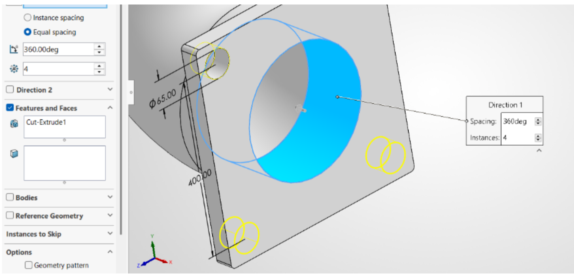
(a) 2D sketch of the base with dimensions



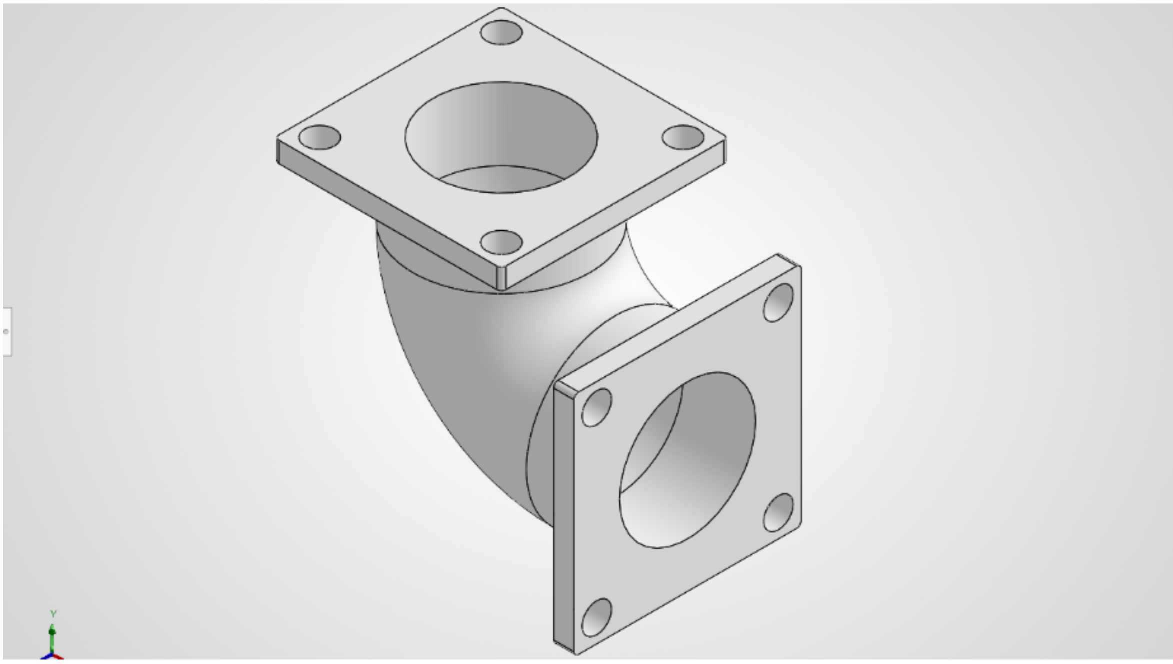
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

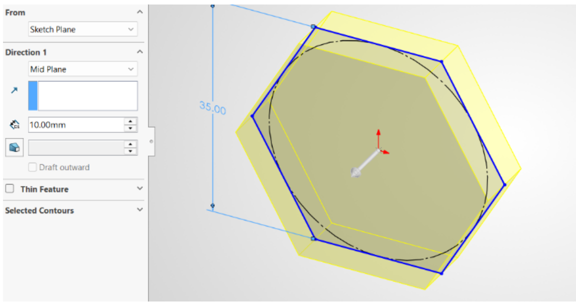


(e) Final result

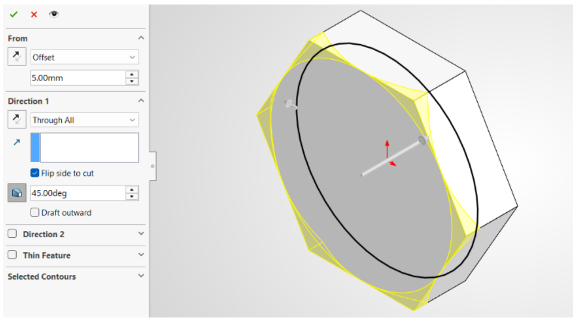
Figure 1.3: Snapshots while Designing Object 3 in Solidworks

1.7 Object 4

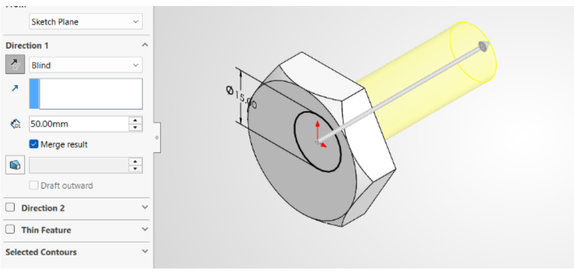
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



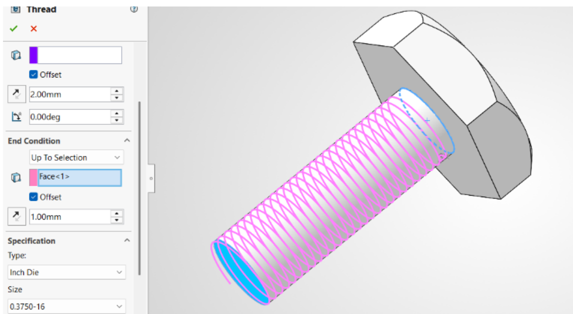
(a) 2D sketch of the base with dimensions



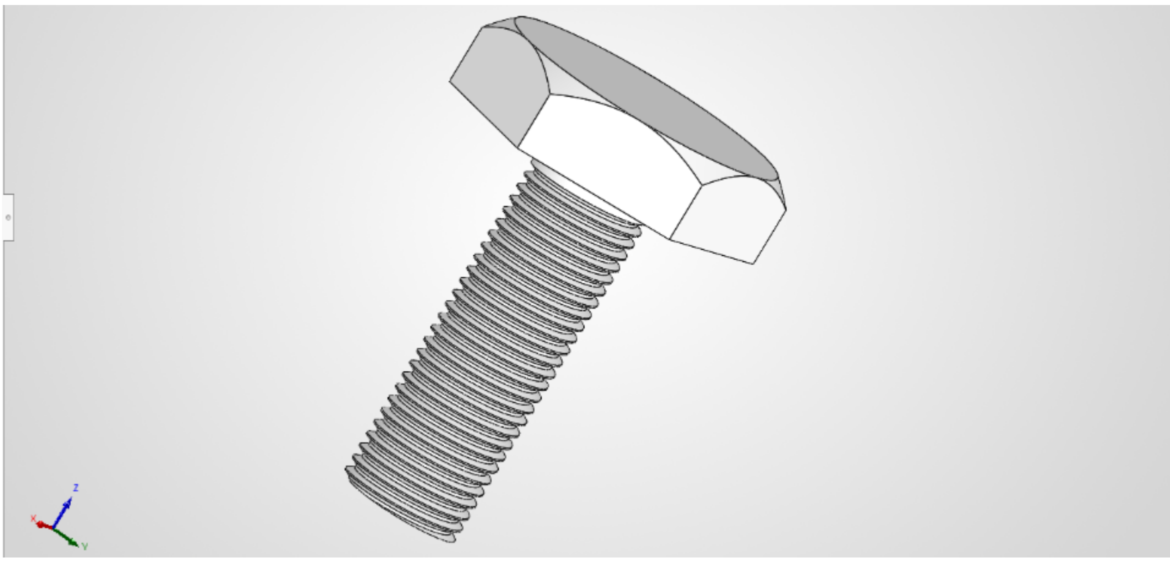
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

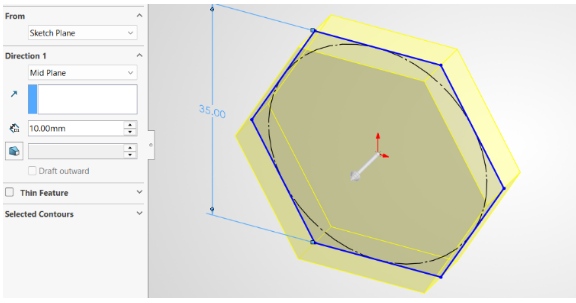


(e) Final result

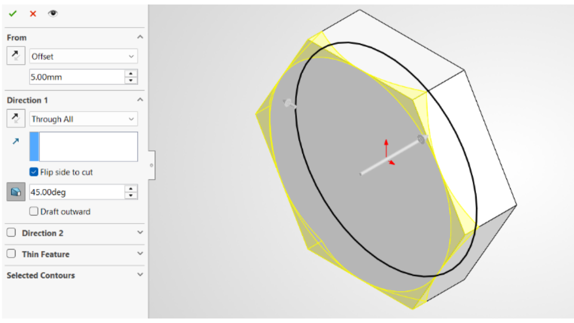
Figure 1.4: Snapshots while Designing Object 4 in Solidworks

1.8 Object 5

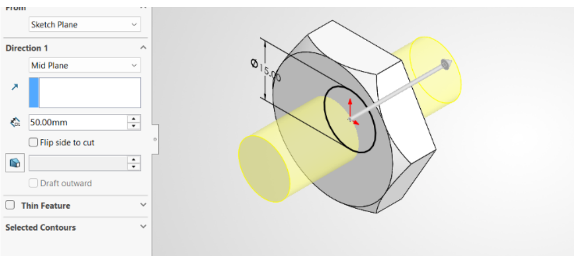
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



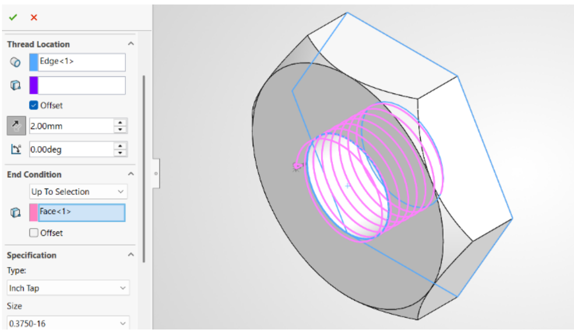
(a) 2D sketch of the base with dimensions



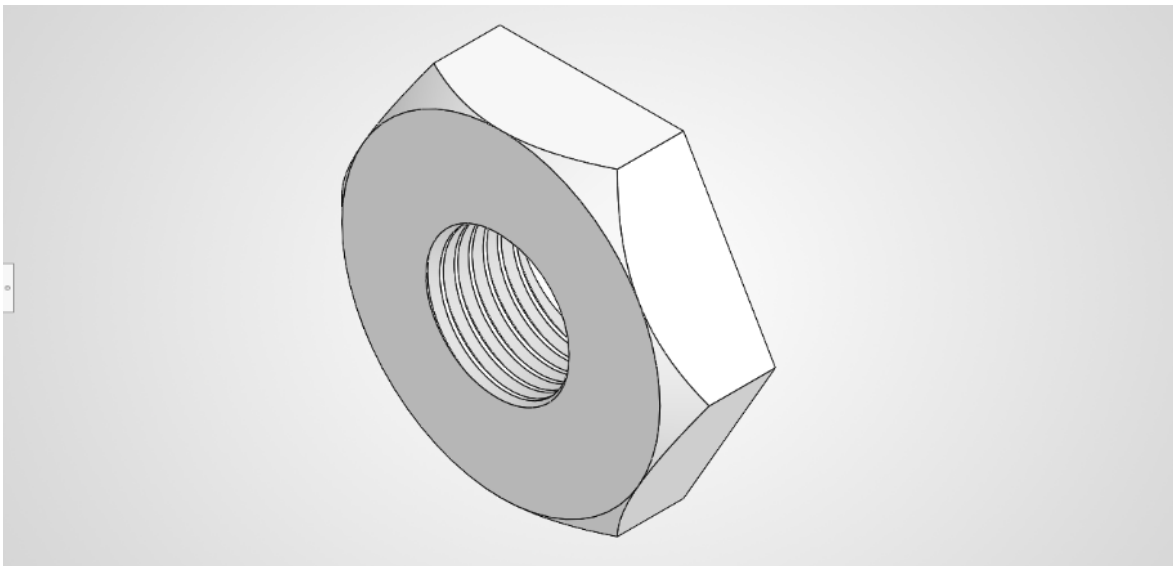
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

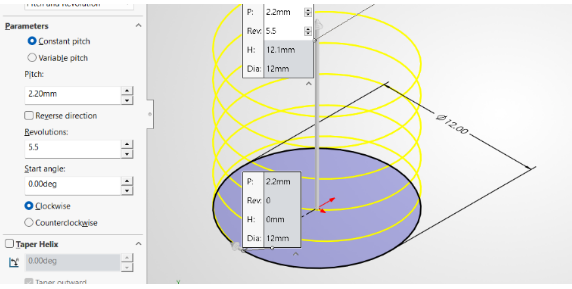


(e) Final result

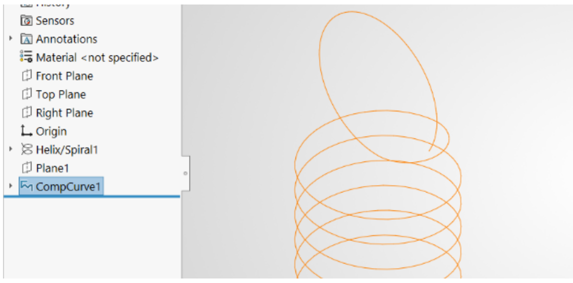
Figure 1.5: Snapshots while Designing Object 5 in Solidworks

1.9 Object 6

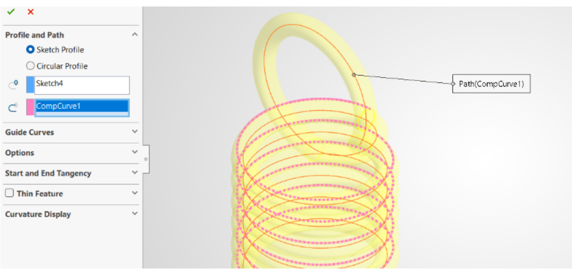
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



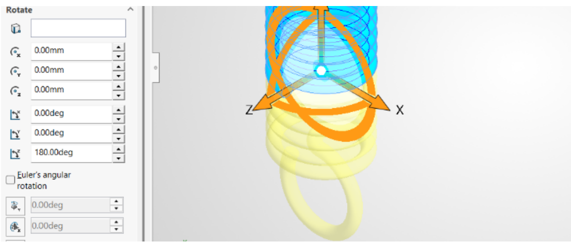
(a) 2D sketch of the base with dimensions



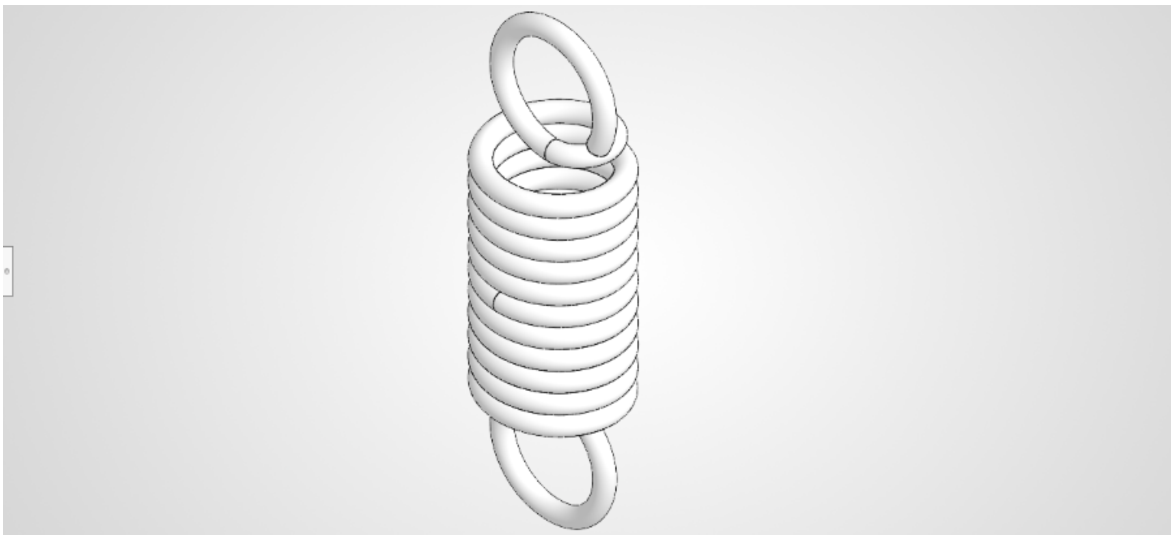
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

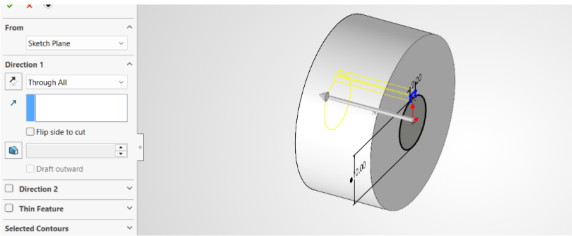


(e) Final result

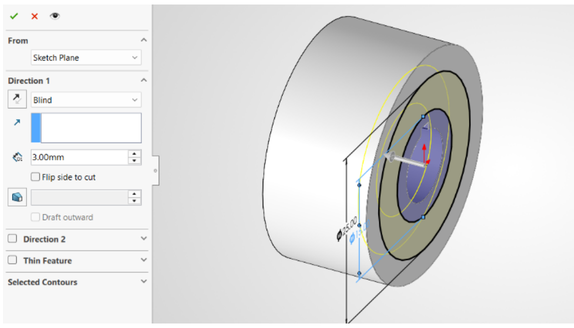
Figure 1.6: Snapshots while Designing Object 6 in Solidworks

1.10 Object 7

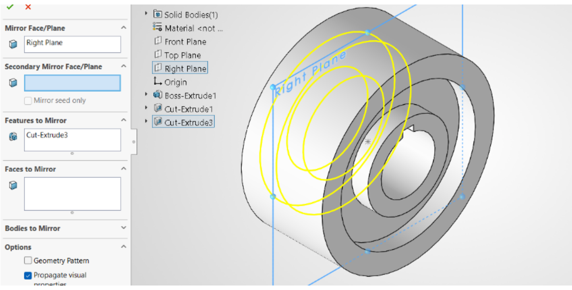
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



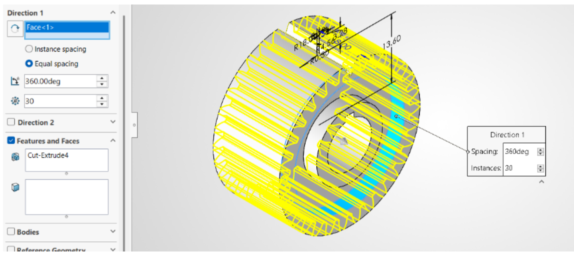
(a) 2D sketch of the base with dimensions



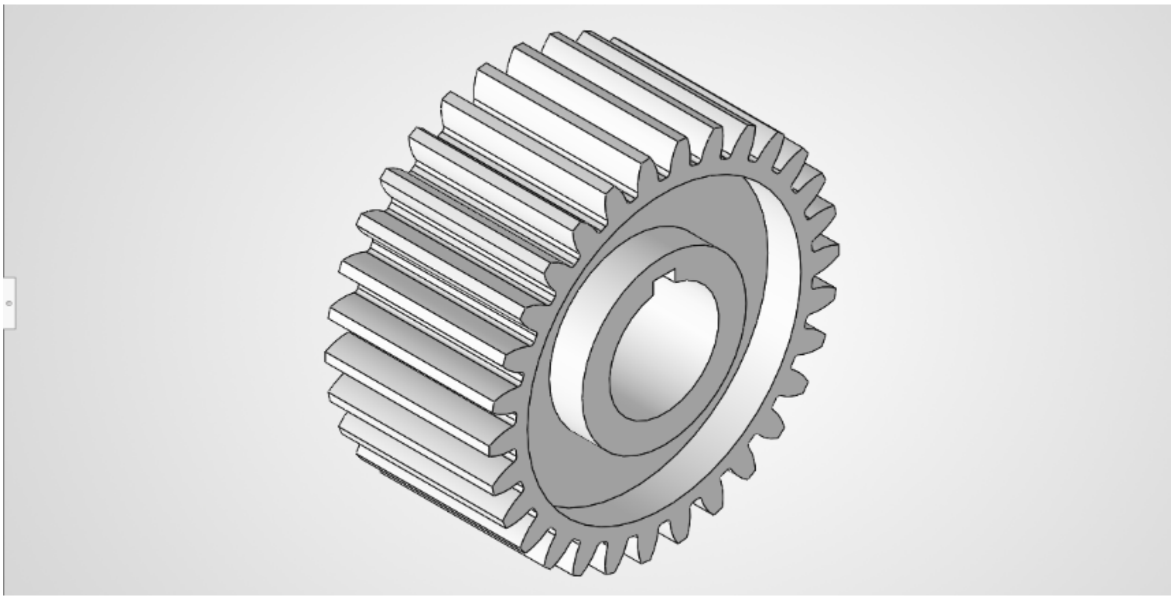
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion



(e) Final result

Figure 1.7: Snapshots while Designing Object 7 in Solidworks

1.11 Discussion

The seven parts in this lab required a mix of prismatic and axisymmetric modeling approaches. Parts with rectangular or stepped profiles were modeled primarily with Extruded Boss/Base and subsequent Cut features, while parts exhibiting rotational symmetry were created using Revolved Boss/Base where appropriate.

Key observations:

- Sketch definition: Parts that began with fully constrained sketches avoided rebuild errors and made downstream edits straightforward [1]. - Feature ordering: Creating primary solids first (bosses/revolves) and applying secondary details (fillets, chamfers, cuts) afterwards simplified history-tree management and reduced feature failures [2]. - Use of shell and patterns: Thin-walled geometry was efficiently produced with ‘Shell‘, and repetitive features used ‘Linear/Pattern‘ to keep the model concise and parametric. - Troubleshooting: Common issues included under-defined sketch entities and ambiguous references for revolve axes; naming sketches and reference geometry reduced ambiguity and rebuild errors [3].

The sectioned write-ups for each object describe the chosen workflow, crucial dimensions, and how each feature contributed to the final part. These notes serve both as reproducible instructions and as a basis for design reviews.

1.12 Conclusion

This lab achieved its objectives by guiding students through the parametric modeling of seven parts using SolidWorks. Participants practiced creating fully defined sketches, selecting appropriate primary features (Extrude, Revolve), and applying secondary operations (Cuts, Fillets, Chamfers, Shells) to complete the designs. The exercise reinforced best practices for sketching and feature management that improve model robustness and editability [1], [2].

Recommended next steps include exploring assembly modeling with these parts and applying tolerancing/engineering drawing exports to prepare parts for manufacturing workflows [3].

References

[1] M. Lombard, *SolidWorks Bible*, 2022nd. Hoboken, NJ: John Wiley & Sons, 2022.

[2] D. A. Madsen and D. P. Madsen, *Engineering Drawing and Design*, 6th. Clifton Park, NY: Cengage Learning, 2016.

[3] Dassault Systèmes. “3D CAD Software | SOLIDWORKS,” Accessed: Jul. 17, 2026. [Online]. Available: https://www.solidworks.com/product/3d-cad